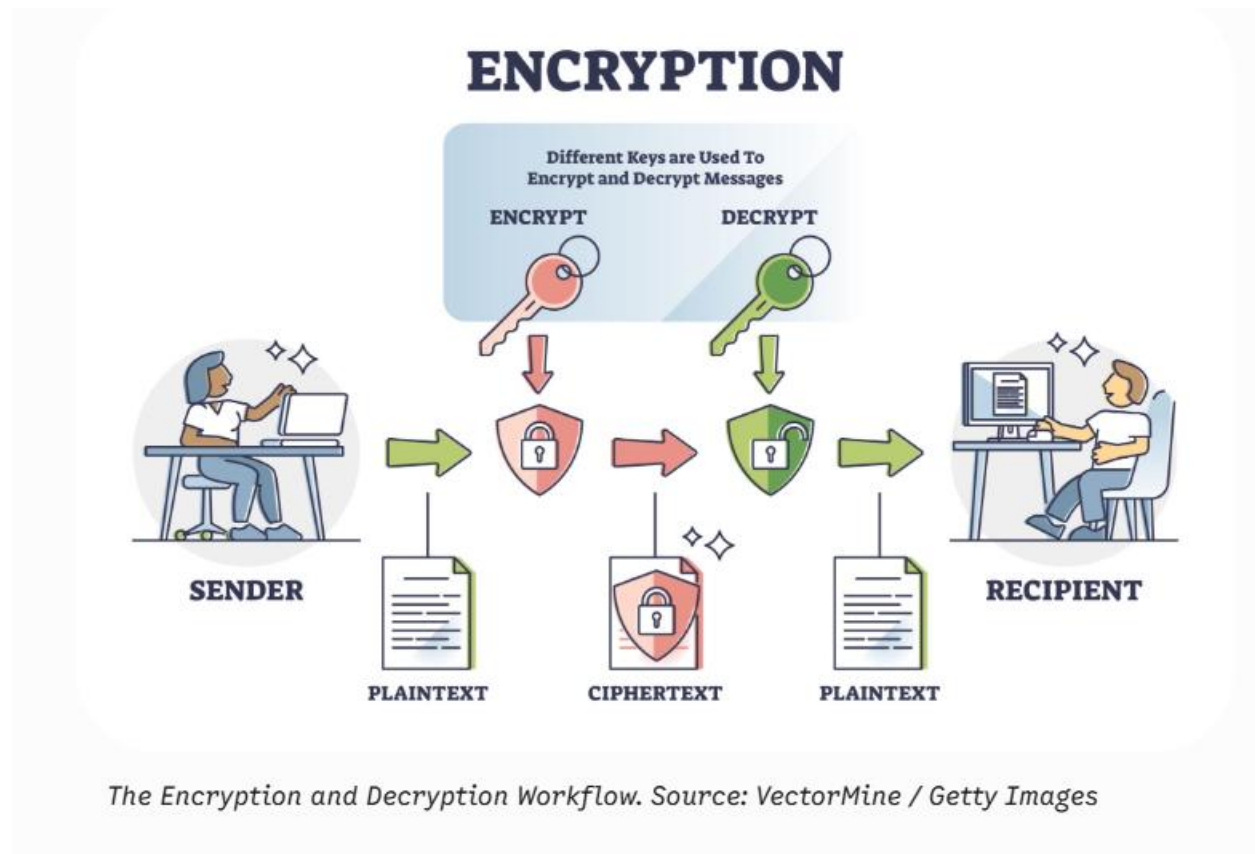




1. Encryption (Locking the Data)

Encryption is the process of translating readable data—called **plaintext**—into an unreadable, chaotic mess of characters called **ciphertext**.

To do this, you need two ingredients:

- **The Cipher (The Math):** The set of rules or the formula used to scramble the data (like AES or RSA).
- **The Key (The Password):** A unique string of bits (numbers and letters) that fills in the blanks of the formula.

Without the key, the math just generates a random, useless block of text.

A Simple Example (The Caesar Cipher): Imagine your formula is just "shift

Modern computer encryption uses this exact same logic, just with math that is exponentially more complex, making it mathematically impossible to guess the original message without the key.

2. Decryption (Unlocking the Data)

Decryption is the exact reverse of encryption. It takes the unreadable **ciphertext** and uses the proper key to convert it back into the original, readable **plaintext**.

If someone tries to read the data without the key, it will just look like gibberish. If they try to decrypt it using the *wrong* key, the math fails, and it either outputs a broken error or entirely different gibberish.

The Core Lifecycle

As shown in the diagram above, the basic flow always follows these exact steps:

1. **Plaintext:** The sender creates a message (e.g., "Hello!").
2. **Encryption:** An algorithm combines the plaintext with an *encryption key* to scramble it.
3. **Ciphertext:** The scrambled message travels across the internet safely. If a hacker intercepts it here, they see nothing but nonsense.
4. **Decryption:** The recipient's device receives the ciphertext and uses a *decryption key* to unscramble it.
5. **Plaintext:** The recipient reads the original message ("Hello!").

Depending on the system, the key used to encrypt and the key used to decrypt might be identical (Symmetric) or they might be a unique, matched pair (Asymmetric). Either way, the fundamental goal remains the same: ensuring data is unreadable while it moves from point A to point B.